ASSESSMENT TOOL 3

TRANSFORMER MECHANICS WORKSHEET

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Tokenize a Sentence

Sentence:

"The cat sat on the mat."

Tokens:

["The", "cat", "sat", "on", "the", "mat", "."]

Explanation:
Tokenization is the process of splitting a sentence into smaller units called **tokens**, which can be words, subwords, or punctuation.

2. Convert Tokens to Token IDs

Token	Token ID (Example)
The	101
cat	205
sat	312
on	156
the	102
mat	478
.	119

Token IDs:

[101, 205, 312, 156, 102, 478, 119]

Explanation:
A tokenizer converts each token into a unique numerical ID that the LLaMA model can process.

3. Explain Embeddings

Definition:

Embeddings are dense numerical vectors that represent the meaning of words or tokens.

Example:

cat → [0.21, -0.54, 0.83, ...]

dog → [0.24, -0.51, 0.80, ...] Since cat and dog have similar meanings, their embeddings are close together in vector space.

Purpose of Embeddings

- **Convert words into numbers**
- **Capture semantic meaning**
- **Help the Transformer understand relationships between words**

4. Label a Transformer EncoderInput Sentence

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Tokenization

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Token IDs

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Embedding Layer

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Positional Encoding

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Multi-Head Self-Attention

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▼

Add & Normalize

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▼

Feed Forward Network

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▼

Add & Normalize

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Encoder Output

5. Identify Query (Q), Key (K), and Value (V)

Suppose the sentence is:

"The cat sat on the mat."

Word	Query	Key	Value
The	Searches information	Matches requests	Stores information
cat	✓	✓	✓
sat	✓	✓	✓
mat	✓	✓	✓

Explanation

- **Query (Q): What the current word is looking for.**
- **Key (K): Used to compare with queries.**
- **Value (V): Contains the actual information passed to the output.**

6. Explain Which Words Receive Higher Attention

Sentence:

"The cat sat on the mat."

When processing the word sat, it pays higher attention to:

- **cat ✓ (Who sat?)**
- **mat ✓ (Where did it sit?)**

It pays lower attention to:

- **The**
- **on**

Attention Example

Word	Attention Weight
cat	0.45
mat	0.35
on	0.10
The	0.05
the	0.05

Conclusion: Important words receive higher attention weights because they contribute more to the sentence meaning.

7. Compare Transformer Models

Feature	BERT	GPT	LLaMA
Architecture	Encoder	Decoder	Decoder
Training	Bidirectional	Left-to-right	Left-to-right
Main Use	Text Understanding	Text Generation	Chatbots & Text Generation
Open Source	Yes	Mostly No	Yes
Efficiency	Medium	High	Very High

Summary

- **BERT: Best for text understanding.**
- **GPT: Best for text generation.**
- **LLaMA: Efficient, open-source, and suitable for research and AI applications.**

8. Complete a Transformer PipelineInput Sentence

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Tokenization

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Token IDs

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Embedding

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Positional Encoding

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Transformer Layers

(Self-Attention + Feed Forward)

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Output Embeddings

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Prediction / Generated Text

9. Draw a Block Diagram

INPUT SENTENCE

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Tokenization

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Token IDs

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Embeddings

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Positional Encoding

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Multi-Head Self-Attention

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Feed Forward Network

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Layer Normalization

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Transformer Output

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Next Token / Final Output

Conclusion

The LLaMA (Large Language Model Meta AI) uses tokenization, embeddings, positional encoding, and Transformer decoder layers with self-attention to understand context and generate meaningful text. The Query, Key, and Value mechanism enables the model to focus on important words, making LLaMA highly effective for natural language understanding and generation.